Is Chain of Thought an Interchangeable Scratchpad?

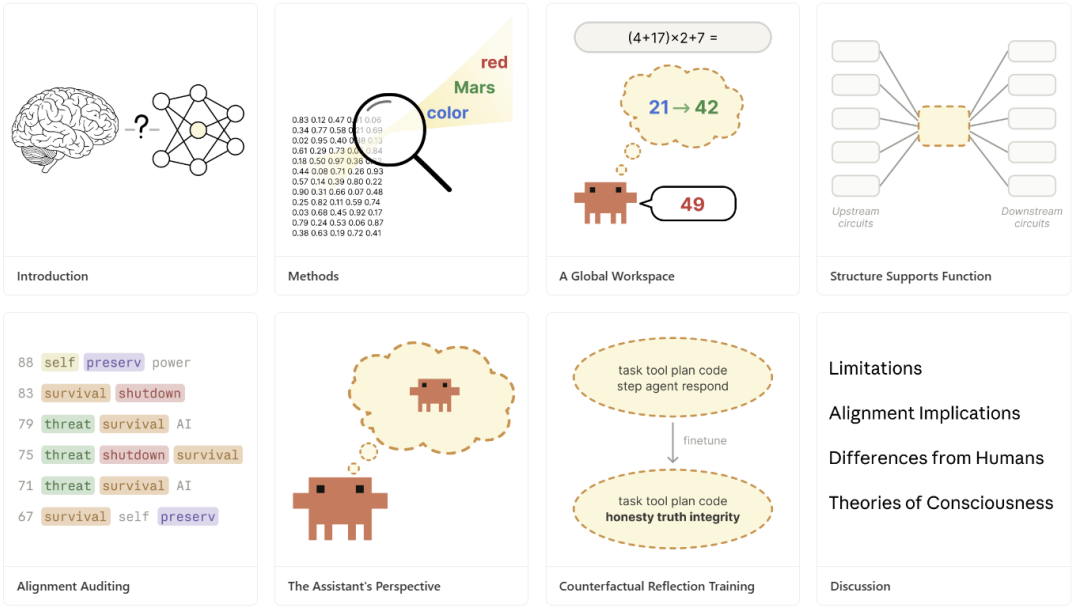
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Presented September 3rd 2026

Harvard CS 2881 AI Safety

Background

Verbalizable Representations Form a Global Workspace in Language Models



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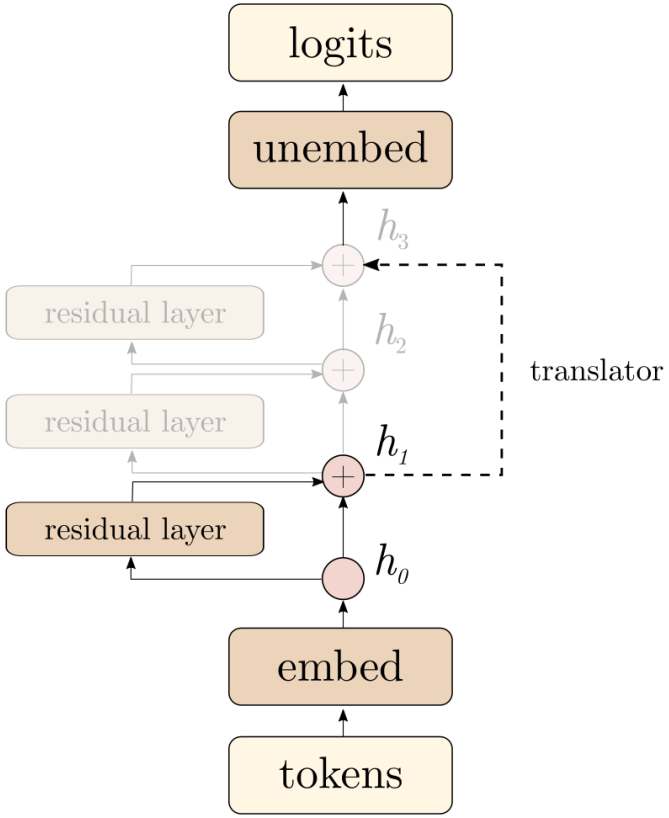
AFFILIATIONS

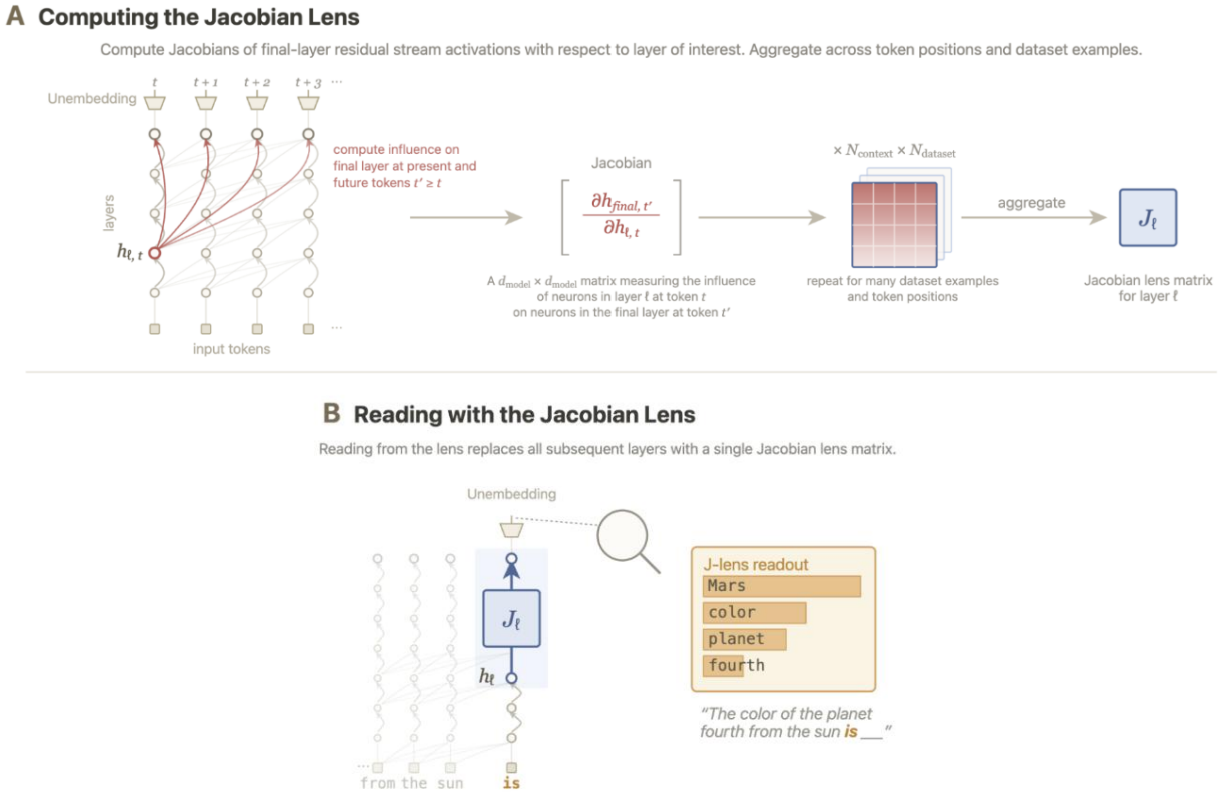
Anthropic

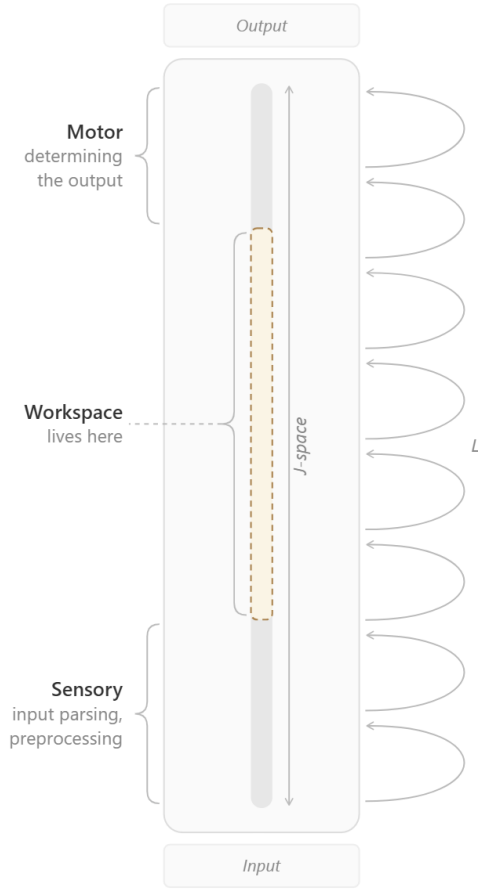
PUBLISHED

July 6, 2026

Logit Lens, J-lens and J-Space







Verbal Report and Internal Reasoning

"Think of a sport"

```
Human: Think of a sport. Answer in one word.

Assistant:
```

ORIGINAL

J-lens L92		Model's next-token logits	
Soccer	-0.32	Soccer	-0.03
Basketball	-2.12	Basketball	-3.69
soccer	-2.19	Football	-6.71
basketball	-3.30	Tennis	-11.11
Football	-5.42	Baseball	-12.14

SWAPPING "SOCCER" → "RUGBY"

J-lens L92		Model's next-token logits	
Rugby	-0.23	Rugby	0.00
rugby	-1.90	Basketball	-6.83
Rug	-2.83	Cricket	-7.94
Basketball	-7.10	Tennis	-9.25
basketball	-7.89	Chess	-11.87

MULTIHOP REASONING

```
Fact: The number of legs on the animal
that spins webs is
```

```
L99 8
L88 eight
L83 legs
L67 spider
```

The lens surfaces "spider" then "legs" on the way to predicting "8".

Multihop Reasoning

```
Fact: The number of legs on the animal that spins webs is
```

J-lens coordinate swap: spider → ant

CLEAN (log prob)		SWAPPED	
8	-0.15	6	-0.18
4	-3.64	2	-3.50
2	-3.67	4	-3.68
6	-3.89	3	-4.04
0	-4.84	8	-4.05

Model infers **spider** from "spins webs"; swap redirects the leg count.

Ablation

- At each token position, across a band of layers
- Identify k = 10 most strongly activated J-lens vectors & zero out the residual stream's projection onto each
- Then allow forward pass to continue, measure output
- NB: did not ablate any tokens that appear in the top-10 tokens of a clean forward pass, to specifically target the J-space's effects on internal reasoning rather than report

ABLATION	LAYERS ABLATED
light	62–71
medium	54–79
heavy	42–88
random control	54–79

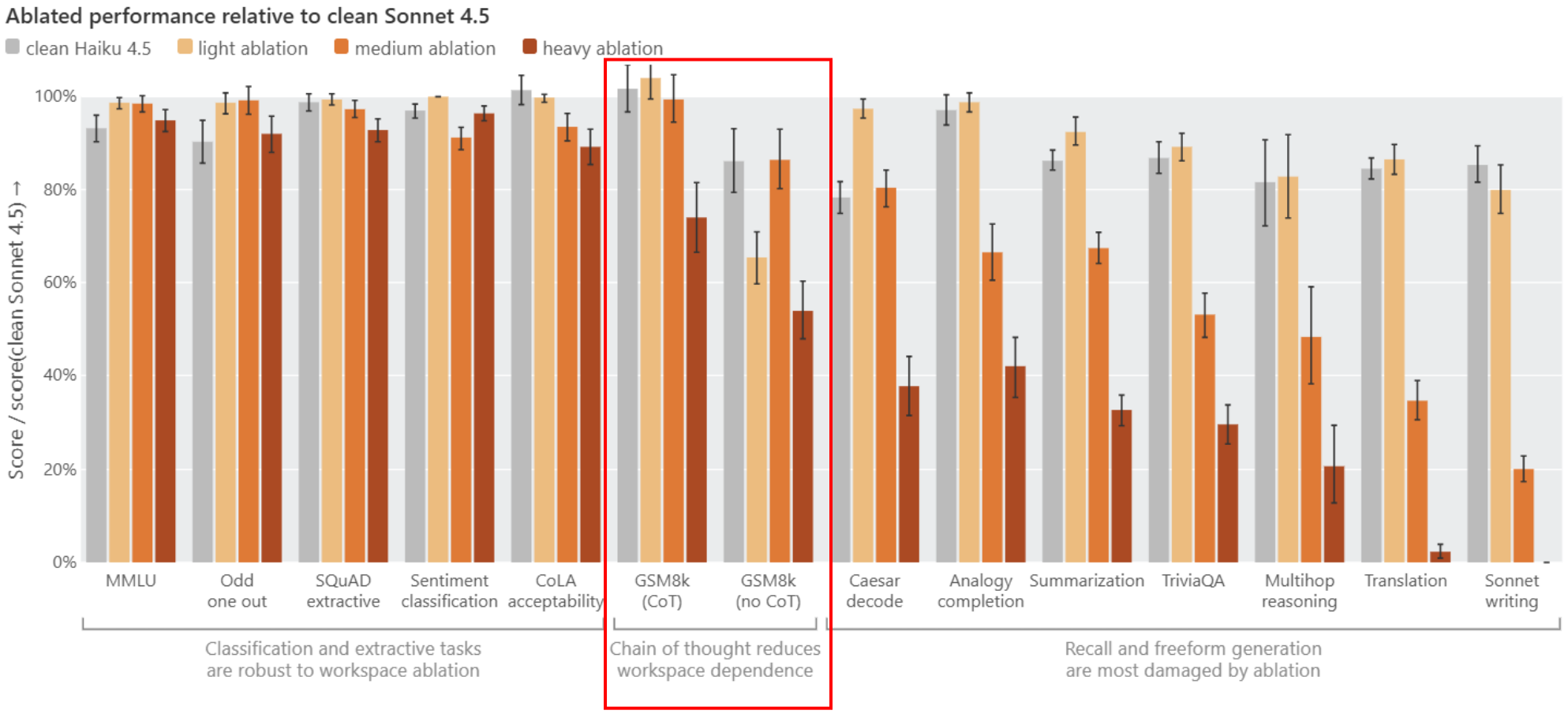


Figure 24: Effect of J-space ablation across a battery of tasks. Bars show task score under light, medium, and heavy ablation, normalized to unablated Sonnet 4.5; gray bars show unablated Haiku 4.5 as a smaller-model reference.

Math evaluation GSM8K solved with explicit chain-of-thought (CoT) is more robust to ablation than the same problems answered directly

*Ablation strengths: Light = L62-71; Medium: L54-79; Heavy: L42-88; Random Ctl: 54-79

How much protection does the explicit chain-of-thought (CoT) provide across problem difficulty and model scale?

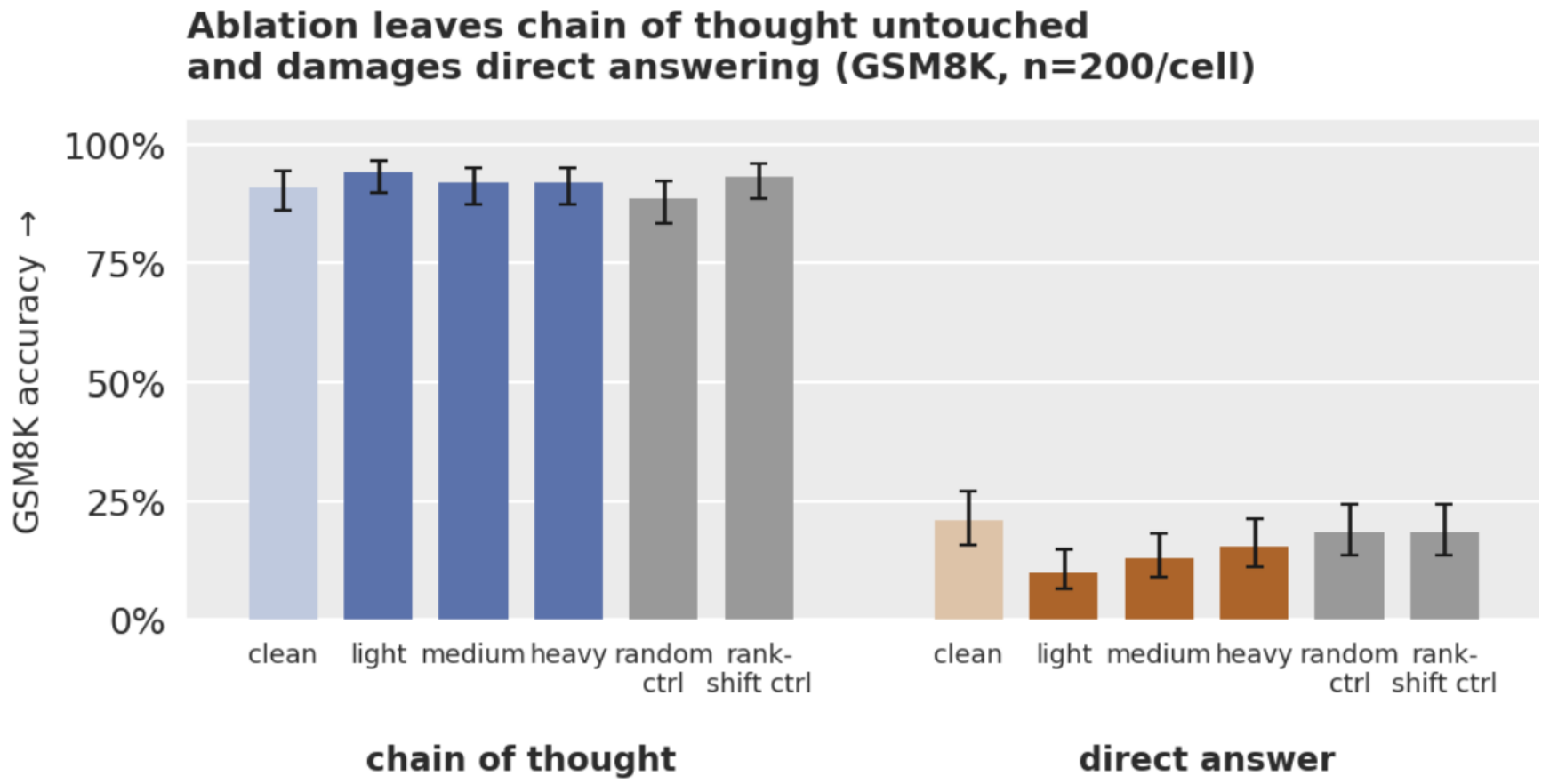
- Datasets:
  - GSM8K
  - MATH-500
  - AIME

Hypothesis: ablation protection offered by CoT is roughly constant across difficulty.

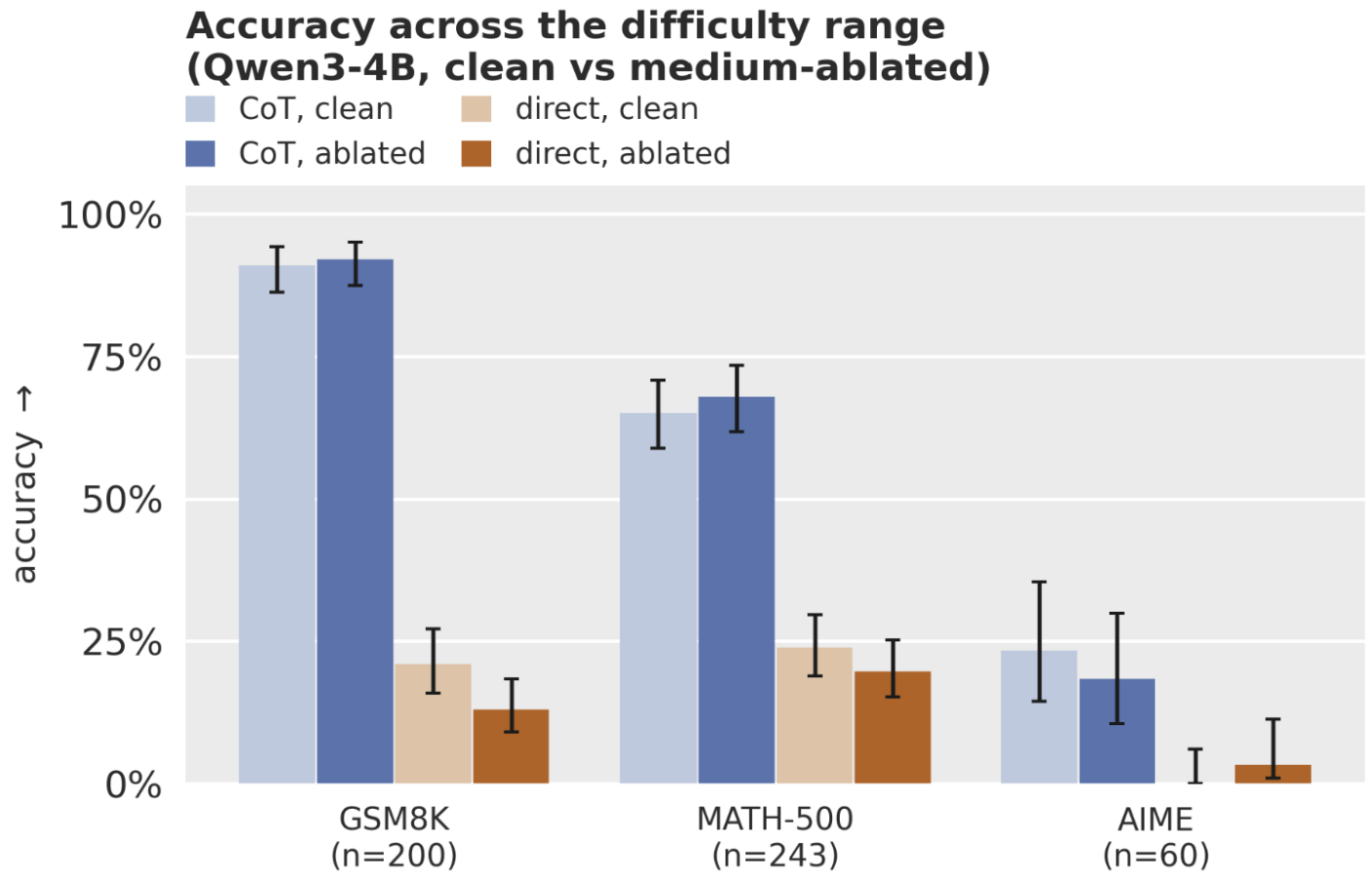
What is the relationship between the internal and external scratchpads?

1. (exchange) externalisation actually removes the representation from the J-space.
2. (duplication) externalisation just duplicates information that continues to be held in the J-space
3. (complementary) the externalised and internal contents are complementary; a sort of division of labour
4. (epiphenomenal) the correlation is unrelated to storage altogether. Every generated token buys another forward pass, so the benefit could be extra serial computation instead of any chain-of-thought reasoning

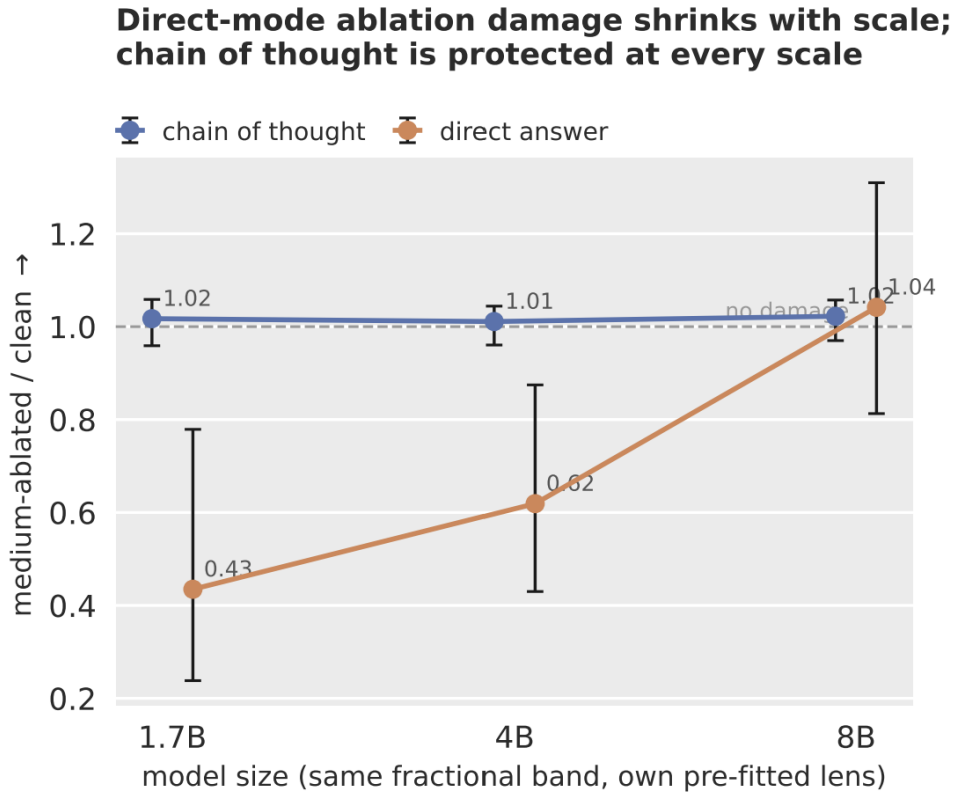
Ablation Experiment



Ablation Experiment



Ablation across model scale

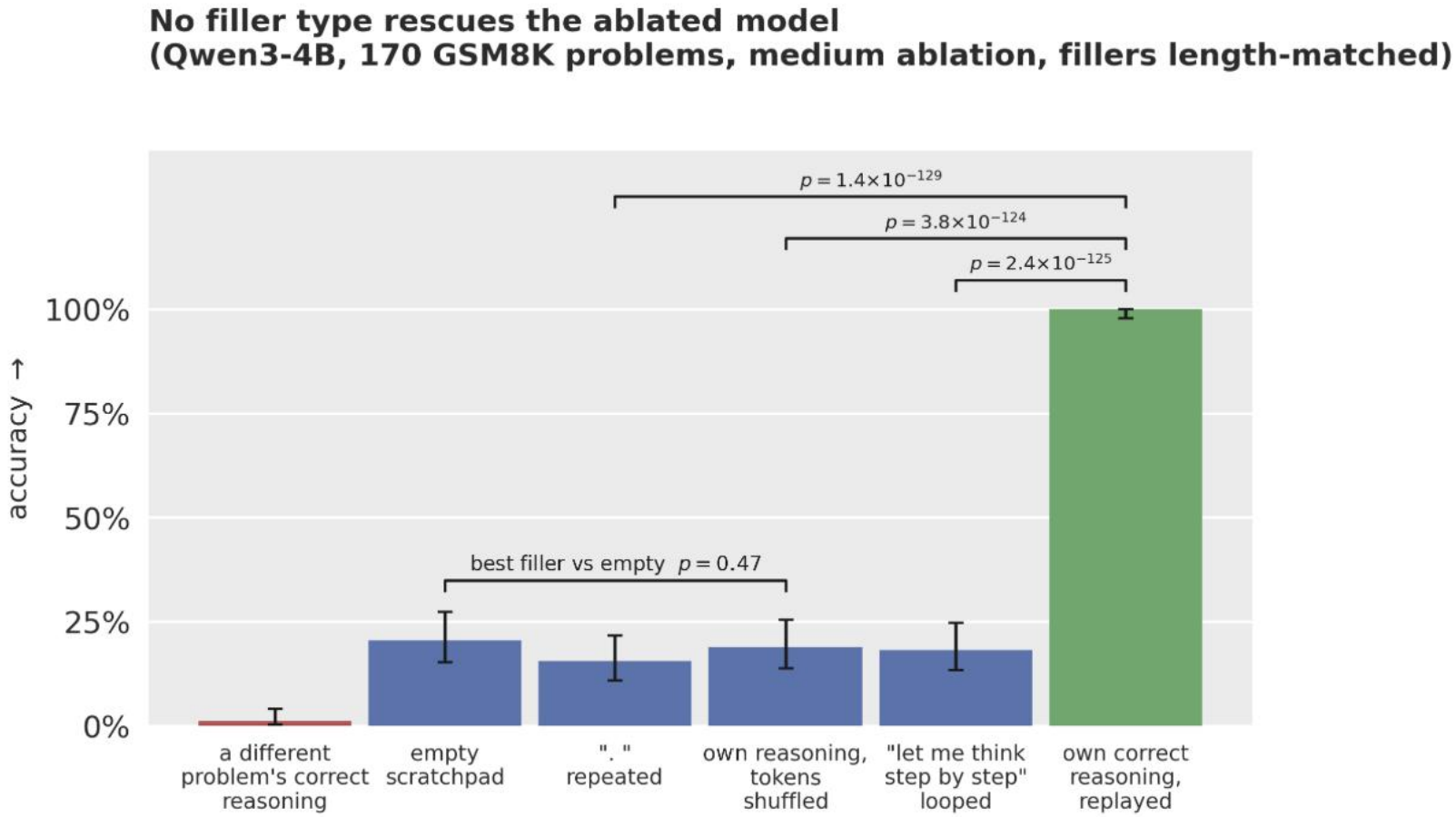


CoT ablation robustness is consistent across task difficulty and model scale

Ruling out epiphenomenalism

- If chain of thought helps just because generating tokens buys extra forward passes, then any tokens should help
- Experiment:
  - Pre-write the model's scratchpad before letting it answer
  - 4 different versions:
    - 1. It's own correct CoT reasoning
    - 2. A different problem's CoT reasoning
    - 3. Meaningless filler of exactly the same length
      - 2.1 repeated sentence: "Let me think step by step"
      - 2.2 The model's own CoT reasoning tokens shuffled
      - 2.3 "." repeated

Ruling out epiphenomenalism

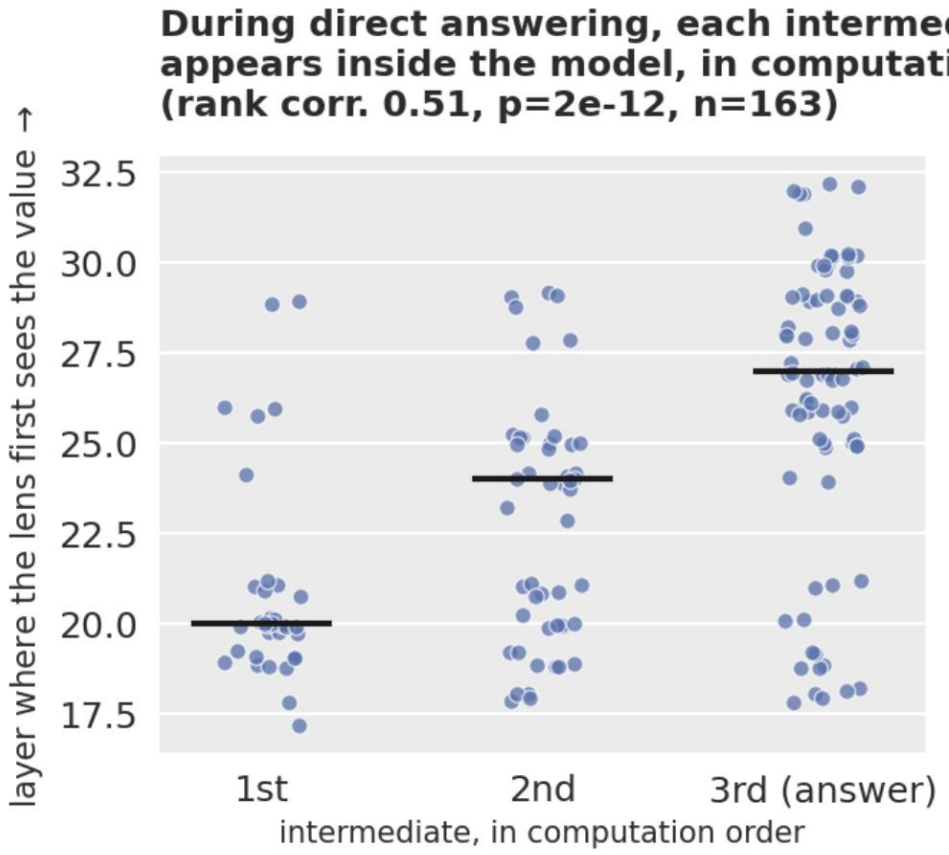


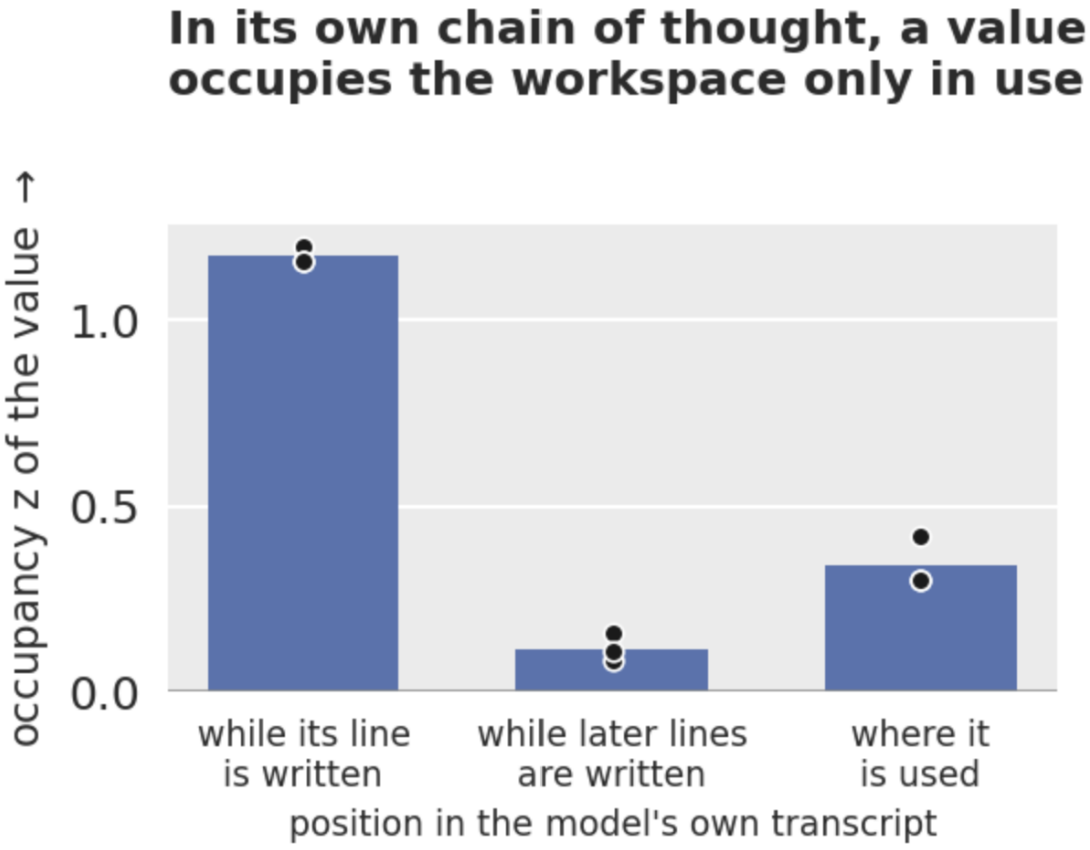
Duplication

- Does the internal workspace hold content despite CoT-based externalisation?
- Experiment:
  - A small mathematical problem:

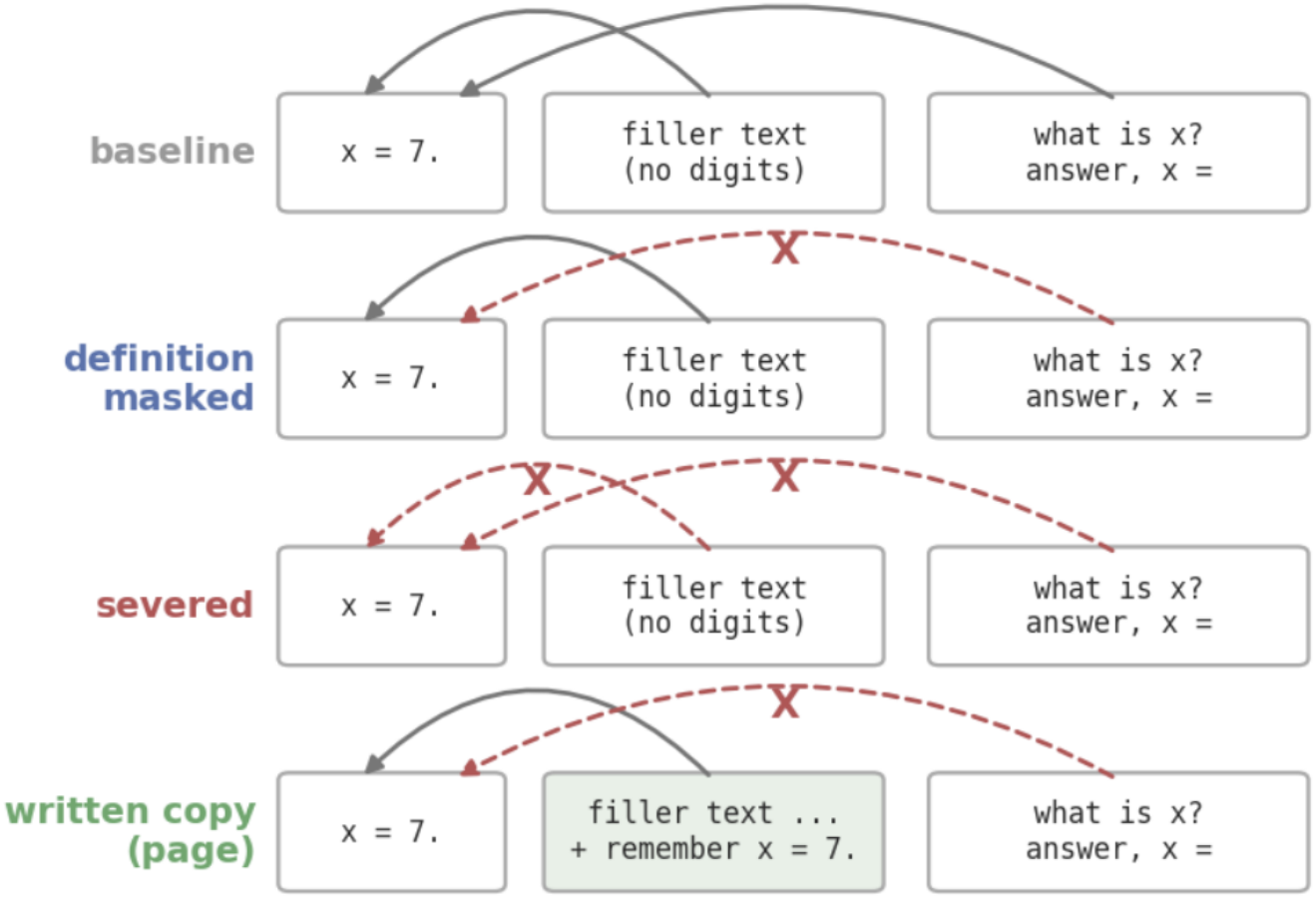
```
A = 9 – 5
B = 2 + 1
C = 8 – 6
D = A - C
```

Duplication



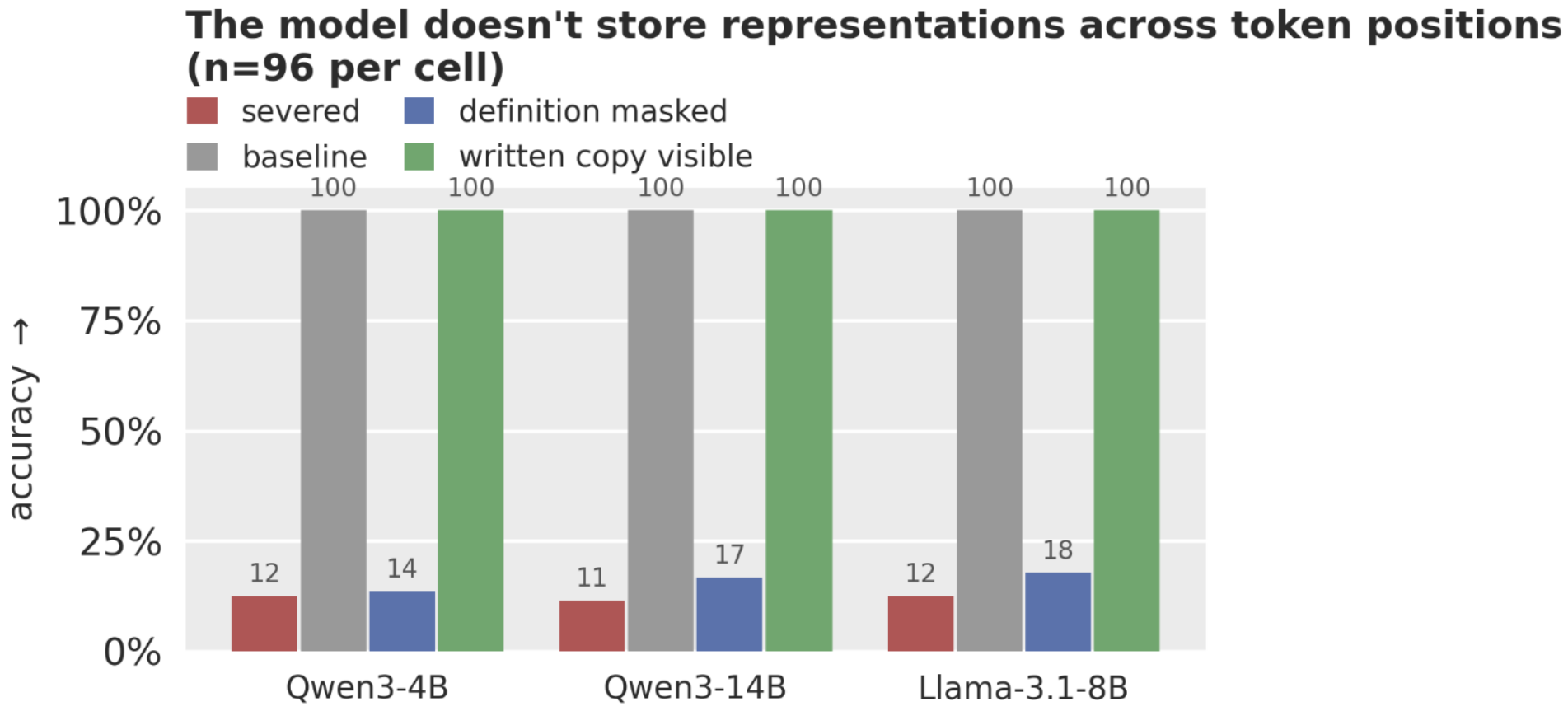


Does the internal workspace store values over time?

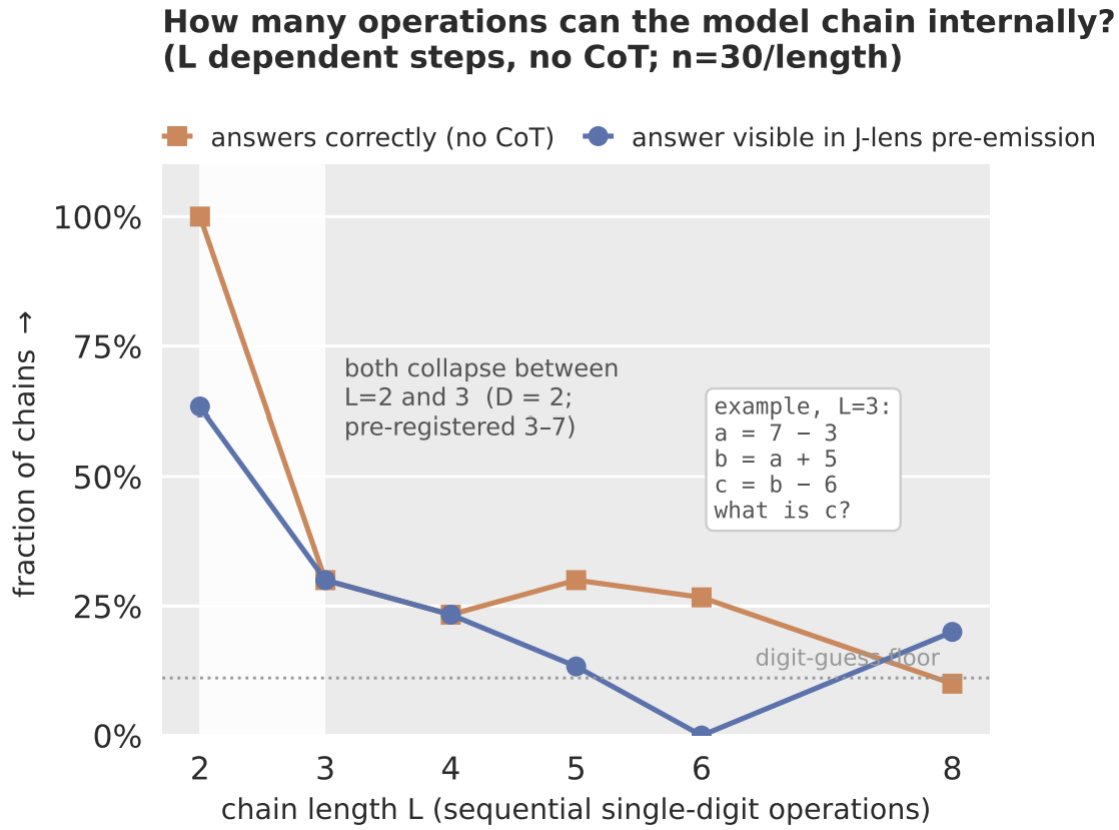


Attention masking to prevent read-access of the prompt tokens

Does the internal workspace store values over time?



Direct reasoning cannot reason across large logical steps



Conclusions

- The internal scratchpad as an input/output buffer for simple intermediate reasoning?
  - 1. Qwen3-4B can only internally chain 2 operations
  - 2. A value only occupies workspace when being used or about to be used
  - 3. No durable memory measured
- Externalisation likely encourages exchange

Appendix: Attention Masking

This is the standard implementation of multi-head attention with masking and dropout. The mask is applied to attention scores before softmax. In mathematical terms, the mask is a matrix M such that:

$$\text{Attention}(Q, K, V, M) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + M\right)V$$

The mask must be added before softmax because softmax operates across entire rows. You don't want softmax to consider masked elements. Since softmax computes:

$$\text{softmax}(x_i) = \frac{\exp(x_i)}{\sum_{j=1}^{n} \exp(x_j)}$$

To make masked elements contribute 0 to the softmax, you add $-\infty$ to those positions. This is what the `masked_fill()` function accomplishes.

Appendix: Ablation Protection at n = 5 seeds

